

YOLO / Object Detection Curriculum

Beginner → Advanced curriculum focused on practical Computer Vision engineering and real-time perception.

LEVEL 1 — OBJECT DETECTION FUNDAMENTALS

1. Object Detection Basics

- Classification vs object detection
- Detection vs segmentation
- Bounding boxes
- Classes
- Confidence scores
- Object localization
- Single-object vs multi-object detection
- Detection pipeline

2. Bounding Boxes

- (x, y, width, height)
- (x1, y1, x2, y2)
- Center-based representation
- Normalized coordinates
- Bounding-box conversion
- Bounding-box visualization

3. IoU — Intersection over Union

- Intersection
- Union
- IoU calculation
- IoU for detection
- IoU thresholds
- IoU limitations

LEVEL 2 — YOLO FUNDAMENTALS

4. What is YOLO?

- You Only Look Once
- One-stage detectors
- Real-time object detection
- YOLO vs two-stage detectors
- YOLO detection pipeline
- Why YOLO is fast

5. YOLO Architecture

- Input
- Backbone
- Neck
- Detection head
- Feature extraction
- Multi-scale detection
- Prediction outputs

6. YOLO Detection Pipeline

- Image → Resize / Letterbox → Preprocessing → YOLO Model → Raw Predictions → Confidence Filtering → NMS → Final Bounding Boxes

LEVEL 3 — YOLO DATASETS

7. Dataset Preparation

- Collecting images
- Image quality
- Class definition
- Dataset structure
- Train / validation / test
- Class distribution
- Dataset leakage
- Data quality checking

8. Image Annotation

- Bounding-box annotation
- Manual annotation
- Annotation tools
- Label formats
- YOLO annotation format
- COCO format
- Pascal VOC format

9. YOLO Label Format

- class_id x_center y_center width height
- Normalized coordinates
- Class IDs
- One .txt file per image
- Multiple objects per image

10. Dataset Configuration

- Dataset YAML
- Training images

- Validation images
- Number of classes
- Class names
- Dataset paths

LEVEL 4 — YOLO TRAINING

11. Training Workflow

- Model selection
- Dataset configuration
- Image size
- Batch size
- Epochs
- Training
- Validation
- Checkpoints
- Best model
- Last model

12. Transfer Learning

- Pretrained YOLO models
- Starting from pretrained weights
- Fine-tuning
- Frozen layers
- Full fine-tuning
- Learning rate

13. YOLO Training Parameters

- Epochs
- Batch size
- Image size
- Learning rate
- Weight decay
- Optimizer
- Warmup
- Augmentation parameters

14. Data Augmentation

- Horizontal flip
- Vertical flip
- Rotation
- Scaling
- Translation

- Cropping
- Color augmentation
- Mosaic augmentation
- MixUp

LEVEL 5 — YOLO LOSS & TRAINING MECHANICS

15. Detection Loss

- Bounding-box loss
- Classification loss
- Objectness/confidence loss
- Total detection loss

16. Bounding-Box Losses

- IoU loss
- GIoU
- DIoU
- CIoU
- Why IoU-based losses are used

17. Classification Loss

- Binary classification
- Multi-class classification
- BCE
- Class probabilities

18. Confidence & Objectness

- Object confidence
- Class confidence
- Final detection confidence
- Confidence threshold

LEVEL 6 — POST-PROCESSING

19. Non-Maximum Suppression — NMS

- Why duplicate detections happen
- Confidence filtering
- IoU threshold
- NMS algorithm
- Class-aware NMS
- Class-agnostic NMS

20. Detection Thresholds

- Confidence threshold
- IoU threshold

- Precision vs recall tradeoff
- Threshold tuning

LEVEL 7 — YOLO EVALUATION

21. Detection Metrics

- Precision
- Recall
- F1 score
- Confusion matrix
- True Positive
- False Positive
- False Negative

22. mAP

- AP
- mAP
- mAP@50
- mAP@50:95
- Precision-recall curve
- Per-class AP

23. Detection Error Analysis

- Missed objects
- False detections
- Wrong class
- Poor localization
- Small-object failures
- Occlusion
- Difficult backgrounds

LEVEL 8 — MODERN YOLO WORKFLOW

24. Ultralytics YOLO

- Installation
- CLI
- Python API
- Model loading
- Prediction
- Training
- Validation
- Export

25. Inference

- Image inference
- Video inference
- Webcam inference
- Batch inference
- Confidence threshold
- Visualization
- Saving results

26. Custom Object Detection

- Collect Dataset
- Annotate
- Prepare YOLO Dataset
- Train
- Validate
- Analyze Errors
- Improve Dataset
- Retrain
- Export
- Deploy

LEVEL 9 — YOLO ADVANCED DETECTION

27. Small Object Detection

- Why small objects are difficult
- Resolution
- Feature maps
- Multi-scale detection
- Image tiling
- SAHI-style inference

28. Crowded Object Detection

- Overlapping objects
- Occlusion
- NMS limitations
- Confidence tuning
- Dataset quality

29. Custom Training Improvements

- Dataset balancing
- Augmentation tuning
- Image resolution
- Batch-size tuning
- Learning-rate tuning

- Model-size selection
- Error-driven improvement

30. Model Variants

- Nano / tiny models
- Small models
- Medium models
- Large models
- Extra-large models
- Accuracy ↔ Speed ↔ Memory ↔ Hardware

LEVEL 10 — YOLO TASKS

31. YOLO Classification

- Image classification
- Training
- Validation
- Inference

32. YOLO Segmentation

- Instance segmentation
- Masks
- Detection + segmentation
- Mask visualization

33. YOLO Pose

- Keypoints
- Human pose
- Skeleton
- Keypoint detection

34. YOLO Tracking

- Detection + tracking
- Object IDs
- Video tracking
- Track management
- Integration with tracking algorithms

LEVEL 11 — YOLO + OPENCV

35. YOLO + OpenCV

- OpenCV image input
- Preprocessing
- YOLO inference
- Bounding-box drawing

- Video processing
- Webcam detection

36. Real-Time Detection

- Camera → OpenCV → Frame → Preprocessing → YOLO → Detection → NMS → Tracking → Visualization
- FPS
- Latency
- Throughput

LEVEL 12 — YOLO DEPLOYMENT

37. Model Export

- PyTorch
- ONNX
- TensorRT
- OpenVINO
- CoreML where relevant

38. ONNX Deployment

- Export
- ONNX Runtime
- Input/output handling
- CPU inference
- GPU inference

39. TensorRT

- TensorRT basics
- FP32
- FP16
- INT8
- Engine creation
- Inference optimization

40. Real-Time Optimization

- Input resolution
- Batch size
- FP16
- INT8
- GPU utilization
- Memory transfer
- Preprocessing optimization
- Post-processing optimization
- FPS vs latency

LEVEL 13 — PRODUCTION YOLO

41. Production Pipeline

- Camera/video input
- Preprocessing
- Inference
- Post-processing
- Tracking
- Output
- Logging
- Monitoring

42. API Deployment

- FastAPI
- Image inference API
- Video inference
- Batch inference
- Docker

43. Model Versioning

- Model versions
- Dataset versions
- Configuration
- Experiment tracking
- Reproducibility

44. Monitoring

- Inference latency
- FPS
- GPU utilization
- Detection confidence
- Detection distribution
- Model degradation

LEVEL 14 — YOLO PROJECTS

Beginner — Project 1: Custom Object Detector

- 3–5 classes
- Custom dataset
- Train YOLO
- Evaluate
- Deploy webcam detection

Intermediate — Project 2: Real-Time Object Detection System

- YOLO + OpenCV
- Webcam
- FPS counter
- Detection visualization
- Configurable confidence

Intermediate+ — Project 3: Multi-Object Detection + Tracking

- YOLO
- Tracking
- Object IDs
- Counting
- Entry/exit detection

Advanced — Project 4: Edge AI Object Detection

- YOLO
- ONNX/TensorRT
- NVIDIA GPU/Jetson
- Real-time optimization
- FPS benchmarking

Production-Level — Project 5: Real-Time Perception System

- Camera → OpenCV → YOLO Detection → Tracking → Object Classification → Spatial / Distance Information → Event Detection → API / Dashboard

Priority for your Computer Vision career

Must master: Object Detection → Bounding Boxes → IoU → YOLO Architecture → Dataset/Annotation → Training → NMS → mAP → Custom YOLO → OpenCV Integration → Tracking → ONNX/TensorRT → Real-Time Optimization.

Then expand into: YOLO Segmentation → YOLO Pose → YOLO Tracking → Edge Deployment.